

Devendran Vemula

+1 (520) 460 3974 | devendranvemula@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Software Engineer with early-career experience building clean, scalable data-driven systems. Delivered backend APIs reducing request latency by 25% and a full-stack dashboard improving business tracking accuracy by 20%. Proficient in Python, SQL, Node.js, and AWS. Seeking to apply system design and data engineering skills to deliver reliable, high-performance solutions.

PROFESSIONAL EXPERIENCE

Vertisize IT Solutions (Client: Raaonline) | *Software Engineer*

Dec 2022 - Dec 2023

- Built and maintained responsive user interfaces using React.js, Redux, TypeScript, and Angular, ensuring efficient state management and seamless integration with backend APIs.
- Designed backend APIs using Node.js and FastAPI, improving latency by 25% and integrating with React.js and Angular applications.
- Optimized PostgreSQL queries, joins, and indexes to reduce execution time and enable faster processing for front-end workflows.
- Applied clean code principles and modular architecture across both front-end and backend components to improve maintainability, scalability, and reduce production defects.
- Partnered with cross-functional engineers using JIRA, UML diagrams, and Agile practices to analyze requirements.
- Deployed and monitored services on AWS (CloudWatch, Auto Scaling) and supported application deployments for front-end and backend services, improving system reliability and reducing downtime.

AplanDev | *Full Stack Developer*

Apr 2022 - Dec 2022

- Built a full-stack web application using React.js, Tailwind CSS, Node.js, and SQL, supporting 500+ users.
- Implemented real-time dashboards and dynamic pricing logic, improving business tracking accuracy by 20%.
- Integrated Google OAuth 2.0 and secure REST APIs, reducing authentication and API-related issues by 30%.
- Automated deployments using GitHub Actions CI/CD pipelines and shell scripts, reducing deployment time and errors.
- Conducted full-stack development training sessions for junior students, teaching React, Node.js, REST APIs, and database fundamentals through hands-on labs, which shortened onboarding time and boosted team productivity.

PROJECTS

Meetly – Meeting Collaboration App

- Designed and built an activity-first social platform using React Native to connect users through real-world experiences.
- Implemented geo-fenced feeds recommending nearby events and casual meetups based on user interests, time, and location.
- Developed a Communities Hub and Trip Planner with moderated groups, itineraries, invitations, and budgeting using a NestJS backend.
- Integrated interactive maps and route planning to improve meetup coordination and reduce planning friction.
- Implemented safety and trust features including verified profiles, privacy controls, and community moderation.
- Architected scalable backend APIs and data models using NestJS, subscriptions, paid experiences, events, and city expansion.

GradeLens – AI Auto-Grading Platform

- Built an AI-powered auto-grading system using Retrieval-Augmented Generation (RAG) to evaluate answers against rubrics.
- Implemented document pipelines, including text chunking and embeddings, for precise semantic retrieval during grading.
- Developed high-performance backend services using Python, Django, and FastAPI to power asynchronous grading workflows, automated feedback loops, and interactive student/professor dashboards.
- Optimized Python-based vector search and Django caching mechanisms, achieving a 30% reduction in LLM token consumption and significantly accelerating response times for grading services.

BudgetBuddy – Smart Expense Management App

- Developed a personal finance management web application enabling expense tracking, budgeting, and spending analysis across categories and time periods.
- Implemented Python-based backend services for transaction processing, budget calculations, and financial analytics.
- Designed and optimized a SQL-based relational database, writing efficient queries for aggregations, trends, and reporting.
- Built machine-learning-based expense forecasting models in Python to predict monthly spending and inform financial decisions.
- Deployed a secure and scalable frontend architecture using Firebase Authentication and Firebase Hosting.

EDUCATION

University Of Arizona | *Master of Science, Data Science* (GPA: 3.75)

Jan 2024 - Dec 2025

Anna University (Panimalar) | *Bachelor of Technology, IT* (GPA: 3.8)

Aug 2019 - Apr 2023

SKILLS

- **Frontend:** React.js, Angular, Next.js, Ionic, Tailwind CSS, Bootstrap, Material UI, Chart.js, PWA
- **Backend:** Node.js, Express.js, FastAPI, REST API Design, JWT, OAuth2, WebSockets, Microservices
- **Databases:** PostgreSQL, MongoDB, MySQL, SQL Server, Oracle SQL, pgvector
- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, Java, SQL, HTML5, CSS3
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Azure, Docker, GitHub Actions, CI/CD, Nginx, Terraform
- **AI & Data:** LLMs, RAG, OpenAI API, NLP, Scikit-learn, Pandas, NumPy, Power BI, Tableau
- **Tools & Methodologies:** Git, Jira, Agile/Scrum, Postman, Linux/Unix, System Design, Performance Optimization